

Assignment 4

Task 1: Simplify and implement

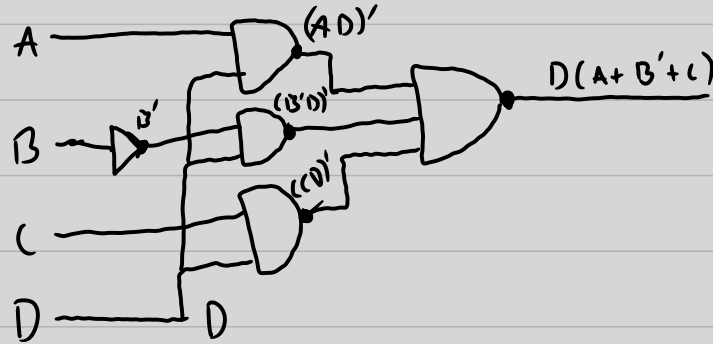
$$a) F(A, B, C, D) = A'B'C'D + CD + AC'D$$

$$= D(A'B'C' + C + AC')$$

* $x + x'y = x + y$ *

$$\left\{ \begin{aligned} &= D(C'(A'B' + A) + C) \\ &= D(C'(A + B') + C) \\ &= D(A + B' + C) \end{aligned} \right.$$

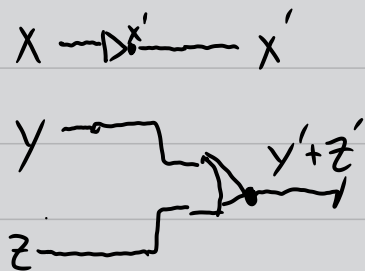
(two level NAND gate circuits)



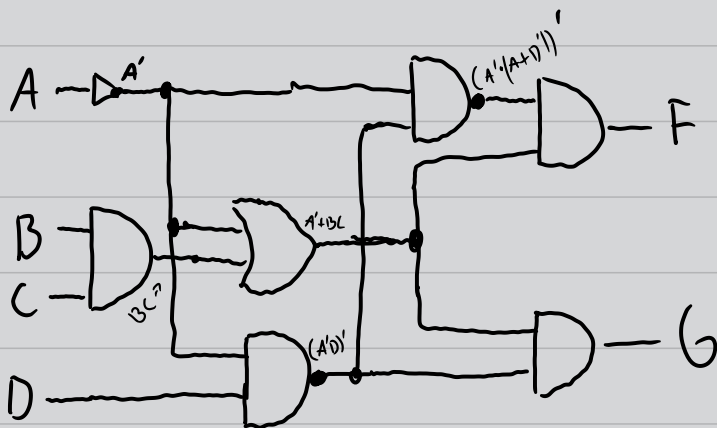
$$b) \bar{F}(x, y, z) = [(x + y)(x + z)]'$$

(two level NOR gate circuits)

$$\begin{aligned} &= ((x + y)(x + z))' \\ &= (x + y)' + (x + z)' \\ &= \underline{x'(y' + z')} \end{aligned}$$



Task 2: Obtain simplified boolean expression for F and G



$$T_1 = BC, T_2 = A' + BC$$

$$\bar{T}_3 = (A' \cdot D)' = A + D'$$

$$G = T_2 \cdot \bar{T}_3 = (A' + BC)(A + D')$$

$$U = A' \cdot \bar{T}_3 = A'(A + D')$$

$$= A'A + A'D' = \underline{A'D'}$$

$$F = U' = (A'D')'$$

$$V = U \cdot T_2$$

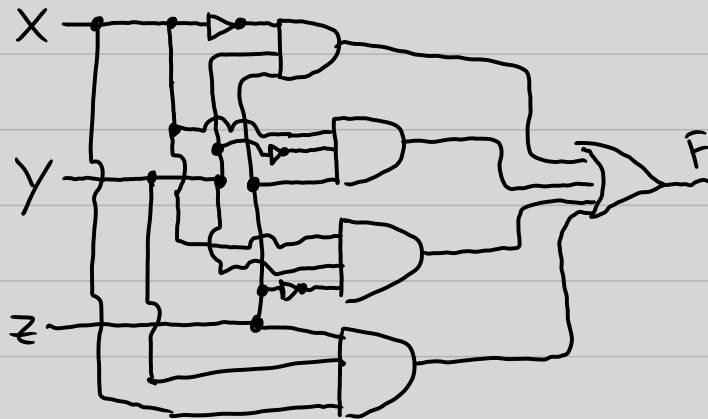
$$= A'D'(A' + BC) = \underline{A'D'}$$

$$F = \underline{\underline{A + D}}$$

* $x \cdot (x + y) = x$ *

Task 3:

x	y	z	$F = x'y'z + xy'z + xyz' + xyz$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1



Task 4:

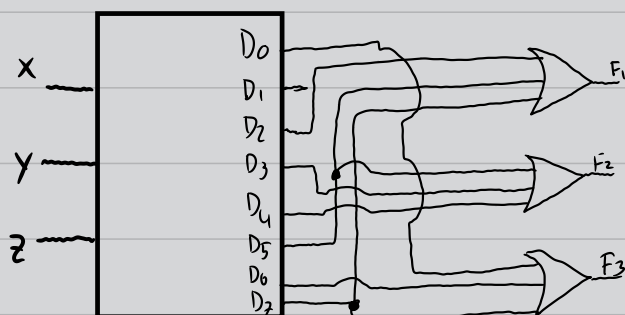
a) $F_1 = x' \cdot y z' + x z$, $F_2 = x \cdot y' \cdot z' + x' \cdot y$, $F_3 = x' \cdot y' \cdot z' + x \cdot y$

$$F_1 = x' \cdot y z' + x z (y + y') = \overbrace{x' y z'}^{D_2 + D_7 + D_5} + \overbrace{x z y + x z y'}^{D_2 + D_7 + D_5}$$

$$F_2 = x y' z' + x' y (z + z') = \overbrace{x y' z'}^{D_5 + D_4 + D_3} + \overbrace{x' y z + x' y z'}^{D_5 + D_4 + D_3}$$

$$F_3 = x' y' z' + x y (z + z') = \overbrace{x' y' z'}^{D_0 + D_7 + D_6} + \overbrace{x y z + x y z'}^{D_0 + D_7 + D_6}$$

	x	y	z	xyz	xy z'	xy' z'	x' y z	x' y z'	x' y' z	x' y' z'
0.	0	0	0	0	0	0	0	0	0	1
1.	0	0	1	0	0	0	0	0	1	0
2.	0	1	0	0	0	0	0	0	0	0
3.	0	1	1	0	0	0	0	1	0	0
4.	1	0	0	0	0	0	1	0	0	0
5.	1	0	1	0	0	1	0	0	0	0
6.	1	1	0	0	1	0	0	0	0	0
7.	1	1	1	1	0	0	0	0	0	0



b) $F_1 = (y' + x) z$,
 $= y' + x z$

$$= y' z (x + x') + x z (y + y')$$

$$= \overbrace{x y' z}^{D_5 + D_1 + D_7} + \overbrace{x' y' z + x y z}^{D_5 + D_1 + D_7}$$

$$F_2 = y' z' + x' y + y z'$$

$$= y' z' (x + x') + x' y (z + z') + y z' (x + x')$$

$$= \overbrace{x y' z' + x' y' z' + x' y z + x' y z'}^{D_4 + D_0 + D_3 + D_2 + D_6} + \overbrace{x y z' + x y z}^{D_4 + D_0 + D_3 + D_2 + D_6}$$

$$F_3 = (x+y)y$$

$$= xy + y$$

$$= xy(z+z') + y(x+x')(z+z')$$

$$= xyz + xyz' + yxz + yxz' + yx'z + yx'z'$$

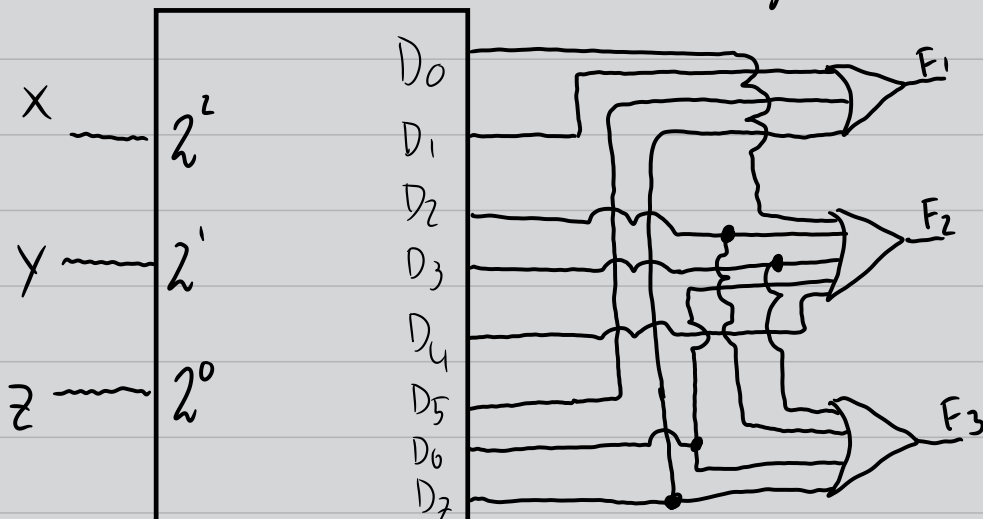
$$= xyz + xyz' + yx'z + yx'z'$$

$$D_7 + D_6 +$$

$$(x'yz + x'y'z')$$

$$D_3 + D_2$$

	X	Y	Z	xyz	xyz'	xy'z'	x'yz	x'y'z'	x'yz'	x'y'z
0.	0	0	0	0	0	0	0	0	0	1
1.	0	0	1	0	0	0	0	0	1	0
2.	0	1	0	0	0	0	0	0	0	0
3.	0	1	1	0	0	0	0	1	0	0
4.	1	0	0	0	0	0	1	0	0	0
5.	1	0	1	0	0	1	0	0	0	0
6.	1	1	0	0	1	0	0	0	0	0
7.	1	1	1	1	0	0	0	0	0	0



Task 5:

D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z	v
1	0	0	0	0	0	0	0	0	0	0	1
x	1	0	0	0	0	0	0	0	0	1	1
x	x	1	0	0	0	0	0	0	1	0	1
x	x	x	1	0	0	0	0	0	1	1	1
x	x	x	x	1	0	0	0	1	0	0	1
x	x	x	x	x	1	0	0	1	0	1	1
x	x	x	x	x	x	1	0	1	1	0	1
x	x	x	x	x	x	x	1	1	1	1	1

$$x = D_4 + D_5 + D_6 + D_7$$

$$y = D_2 + D_3 + D_6 + D_7$$

$$z = D_1 + D_3 + D_5 + D_7$$

If D_2 and D_6 are 1 at the same time. The output will only care about D_6 because of the priority. The output is therefore 1101

Task 6:

$$a) I_1 = I_2 = I_7 = 0, I_3 = I_5 = 1, I_0 = I_4 = D, I_6 = D'$$

A	B	C	D	F_A
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

	$A'B'$	$A'B$	AB	AB'
$C'D'$	0	1	0	0
$C'D$	0	0	1	1
CD	1	0	0	0
CD'	0	1	1	1

$$F = A'B'CD + A'BD' + AC'D + ACD'$$



Alternative answer?:

$$F(A, B, C, D) = \underline{\underline{\sum (1, 6, 7, 9, 10, 11, 12)}}$$

Task 6:

b) $l_1 = l_2 = 0, l_3 = l_7 = 1, l_4 = l_5 = 0, l_0 = l_6 = 0'$

A	B	C	D	F_{10}
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

	$A'B'$	$A'B$	AB	AB'
$C'D'$	1	0	0	0
$C'D$	0	0	1	1
CD	1	0	1	1
CD'	0	1	1	0

$$F = A'B'C'D' + BCD' + B'(CD + AD)$$

Alternative answer?:

$$F(A, B, C, D) = \sum(0, 6, 7, 9, 11, 12, 14, 15)$$

